Head and Neck Embryology

Dr. Gamal Abdein

Dr. Amaal Yousif

Outlines

- Development of the skull.
- Congenital anomalies of the skull.
- Development of the face.
- Congenital anomalies of the face
- Derivatives of the pharyngeal arches.
- Congenital anomalies of the pharyngeal arches.

Development of the Skull

1- Bones of the face (viscerocranium):

- Are derived from the 1st and 2nd pharyngeal arches, from neural crest cells migrated to the arches
- It includes:

A. Cartilaginous viscerocranium :

1. First arch: incus and malleus.
2. Second arch: stapes, styloid process and part of hyoid bone

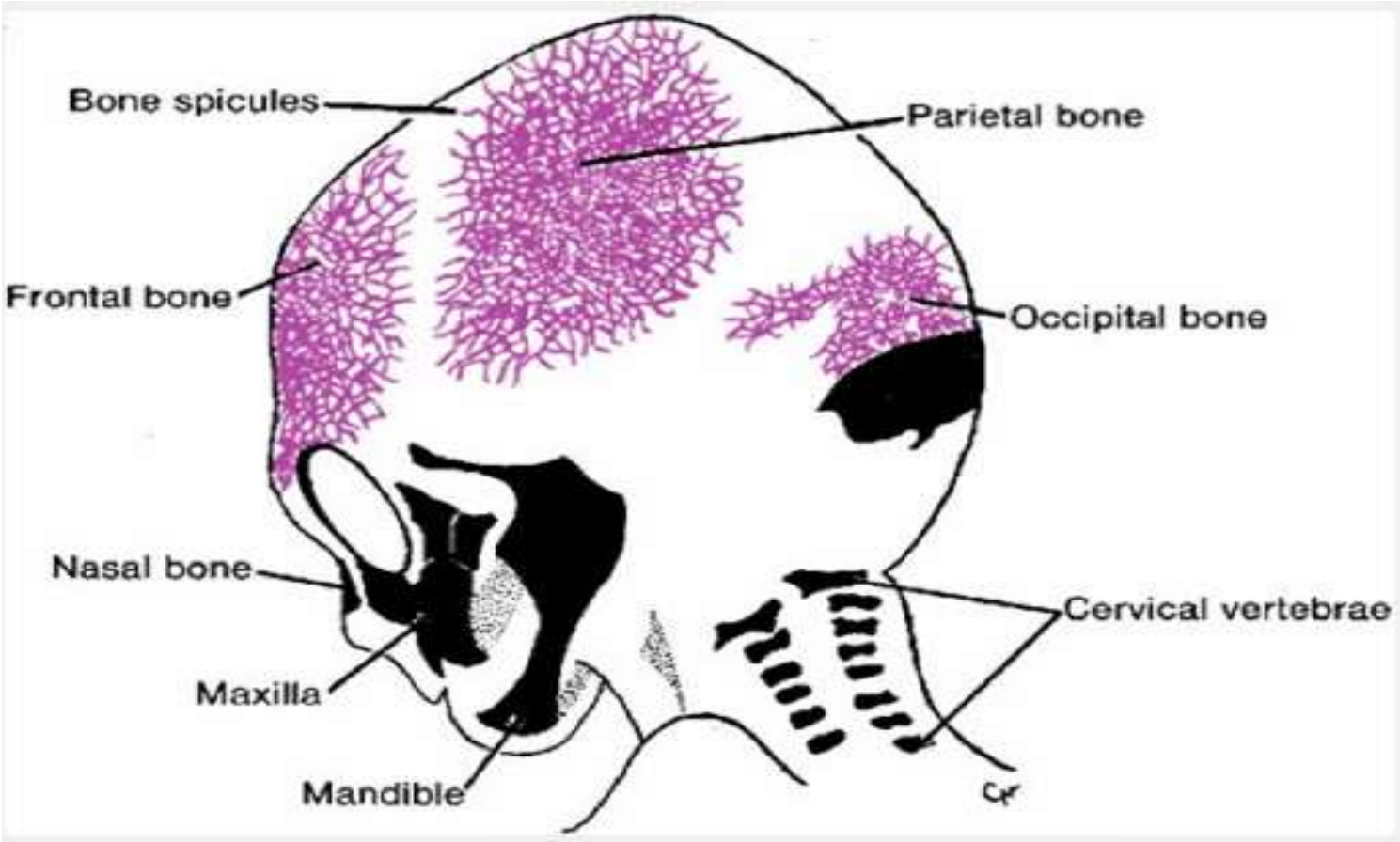
B. Membranous viscerocranium: mandible, maxilla, zygomatic & squamous parts of temporal bone.

Development of the Skull

2- Bones of the cranium (neurocranium):

1) Cartilaginous neurocranium (chondrocranium) → the skull **base**
   - Three middle line cartilages and three lateral cartilages
2) Membranous neurocranium: gives the skull **vault**
   - Centers of ossifications appears in the membrane around the brain → most of frontal, parietal, squamous part of temporal and occipital bones
   - Membranous remnants between bones → sutures and fontanells

Development of the Skull

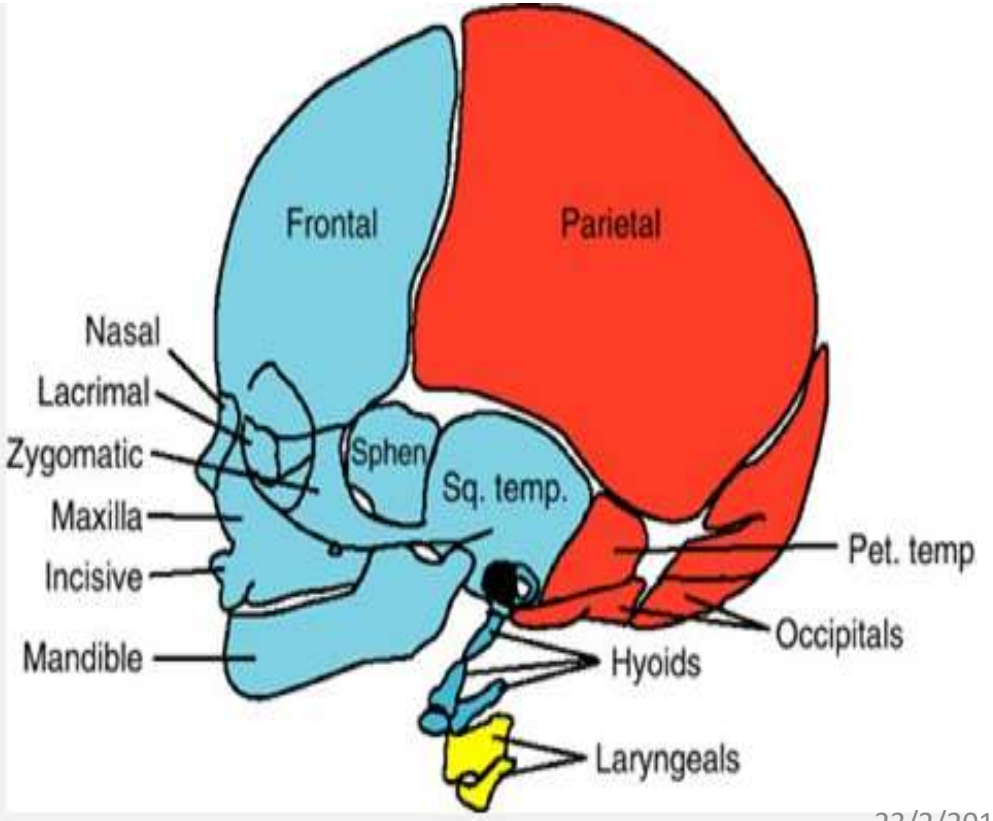


Fetal skull showing centers of ossifications appears in the flat bones of the skull

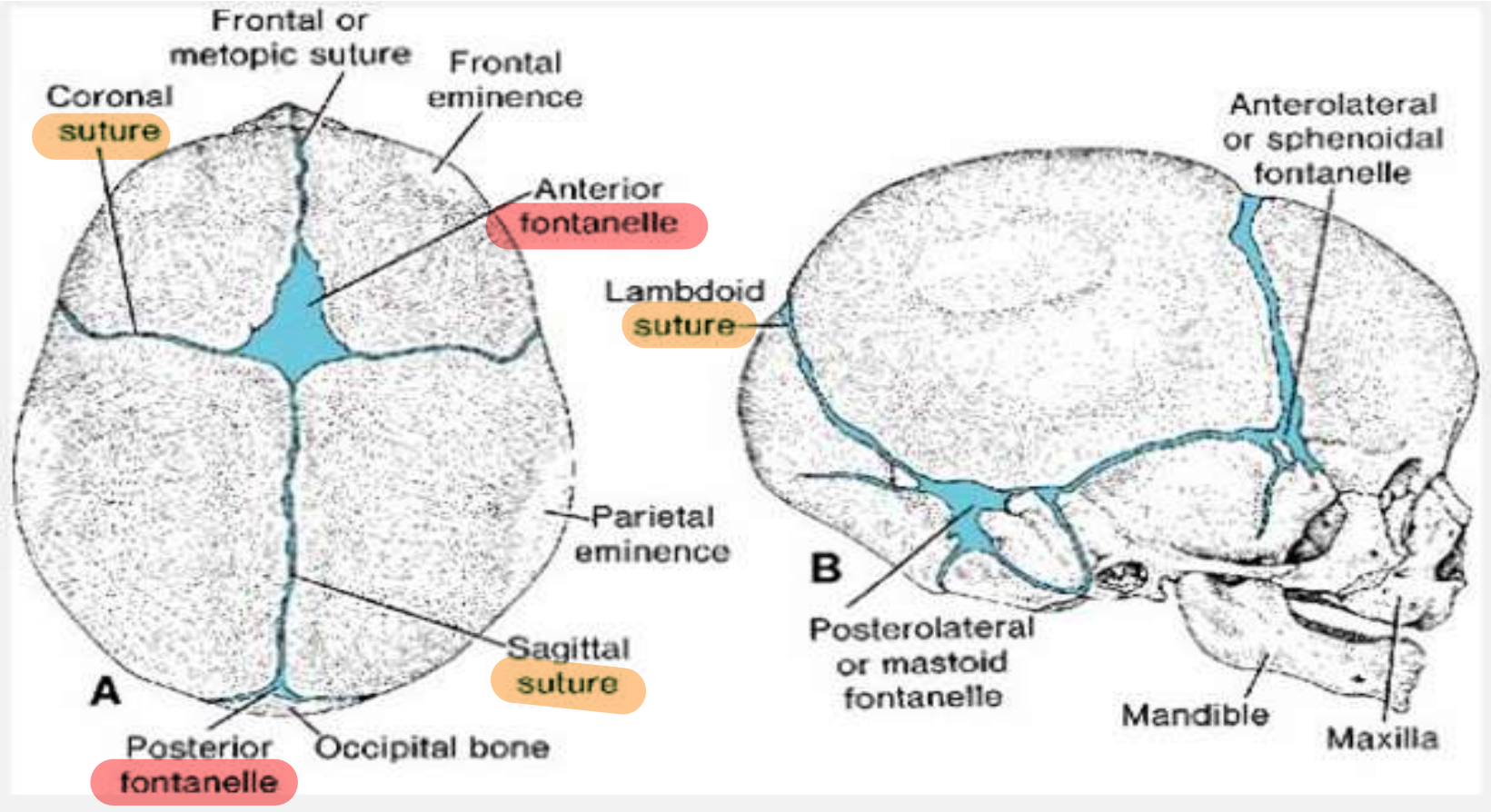
Development of the Skull

Figure showing the skeletal structure of the head and face.

- **Red** from paraxial mesoderm.
- **Blue** from migrating neural crest cells.
- Yellow from lateral plate mesoderm.

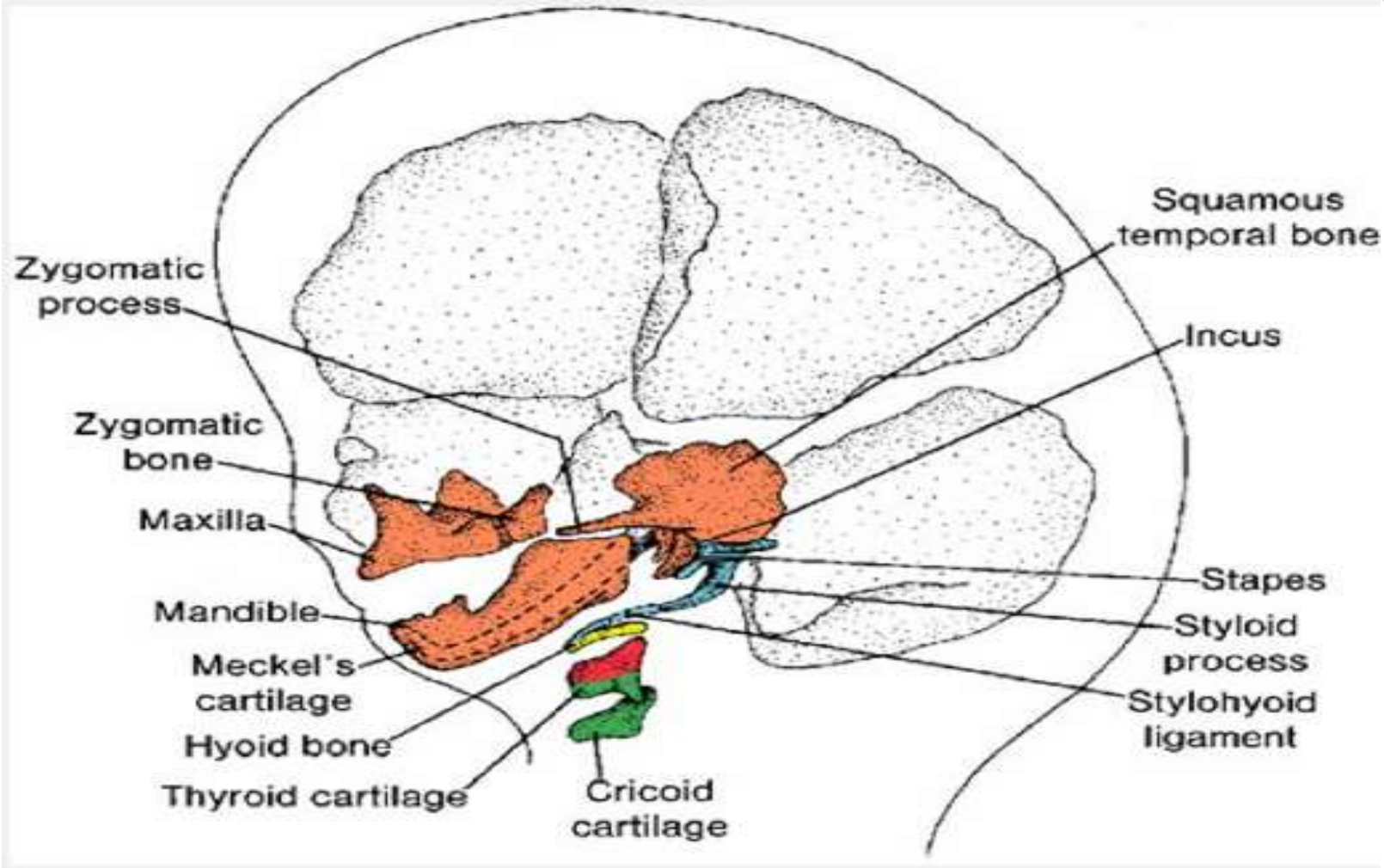


Development of the Skull



Skull of newborn showing sutures and fontanells.

Development of the Skull



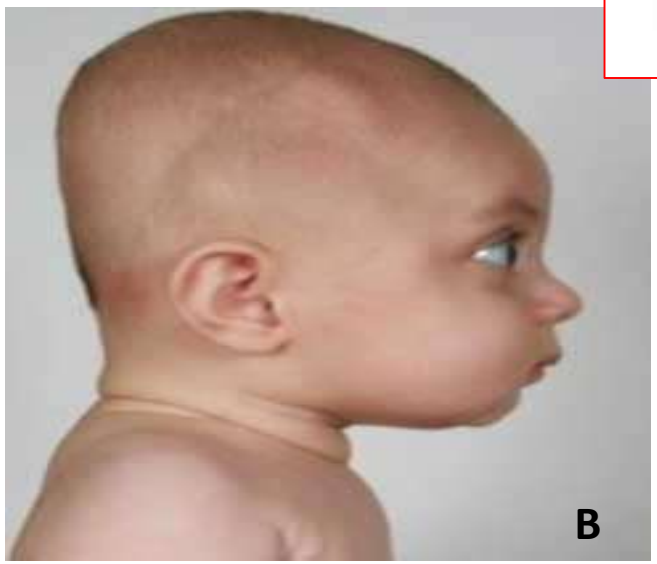
Fetal skull showing the derivatives of the arches enter in formation of the bones of the face

Congenital Anomalies of Skull

1. **Acrania** : Abscent cranial vault . If associated with partial absence of the brain it is **anencephaly** .
2. **Craniostenosis** : Premature closure of the cranial sutures, resulting in malformation of the skull.
   A. Scaphocephaly
   B. Oxycephaly (Acrocephaly)



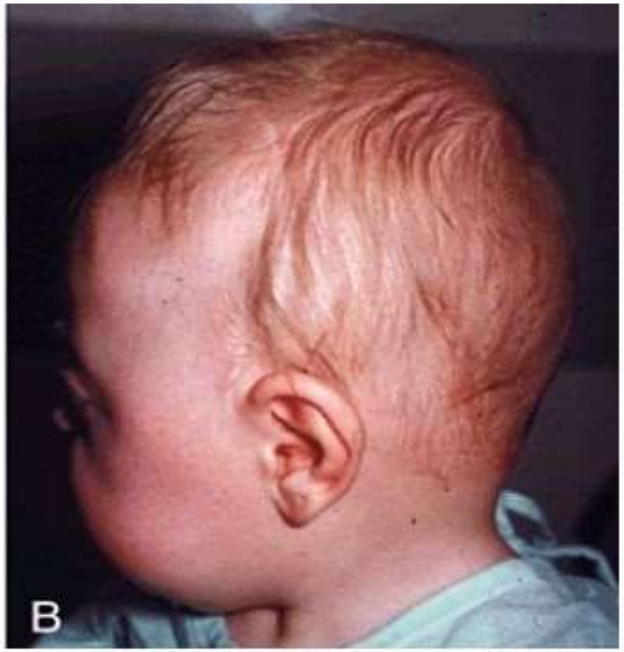


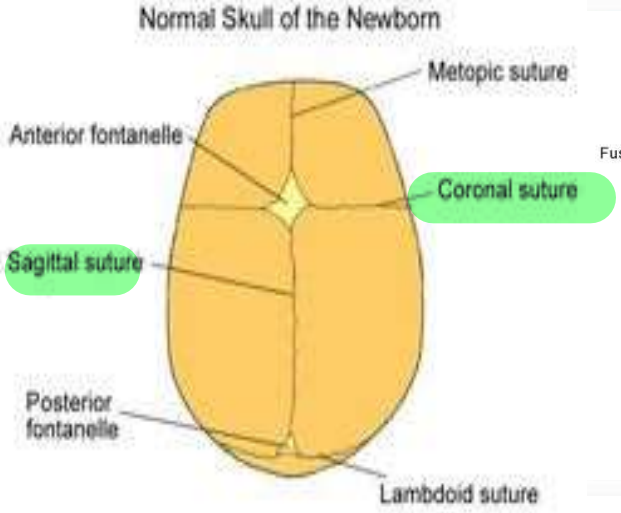


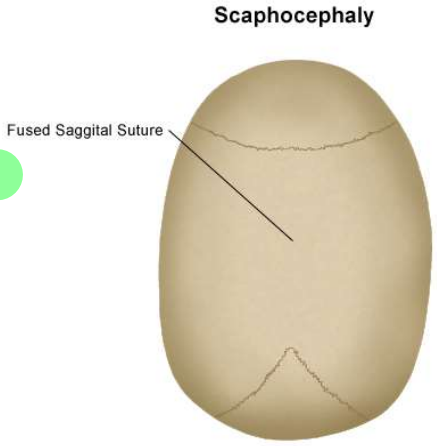
- **Craniosynostosis :**
- Caused by premature closure of one or more sutures.
- **Scaphocephaly:**

  Early closure of the sagittal suture
- **Acrocephaly or tower skull:** Premature closure of the coronal suture.









Congenital Anomalies of the Skull

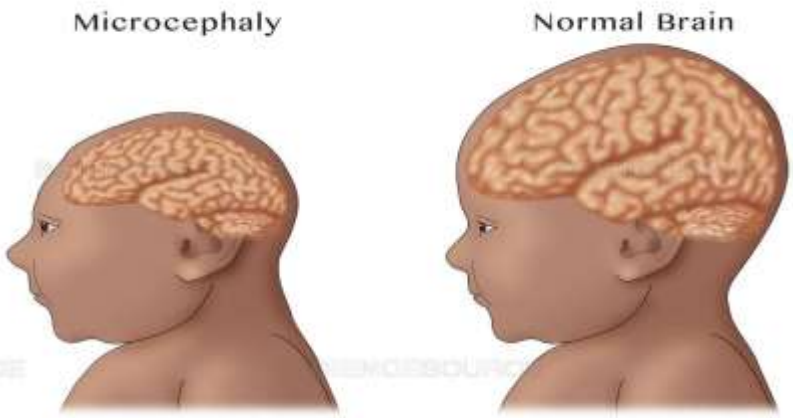
c- Plageocephaly:

Flattening to one side at the back of the head.



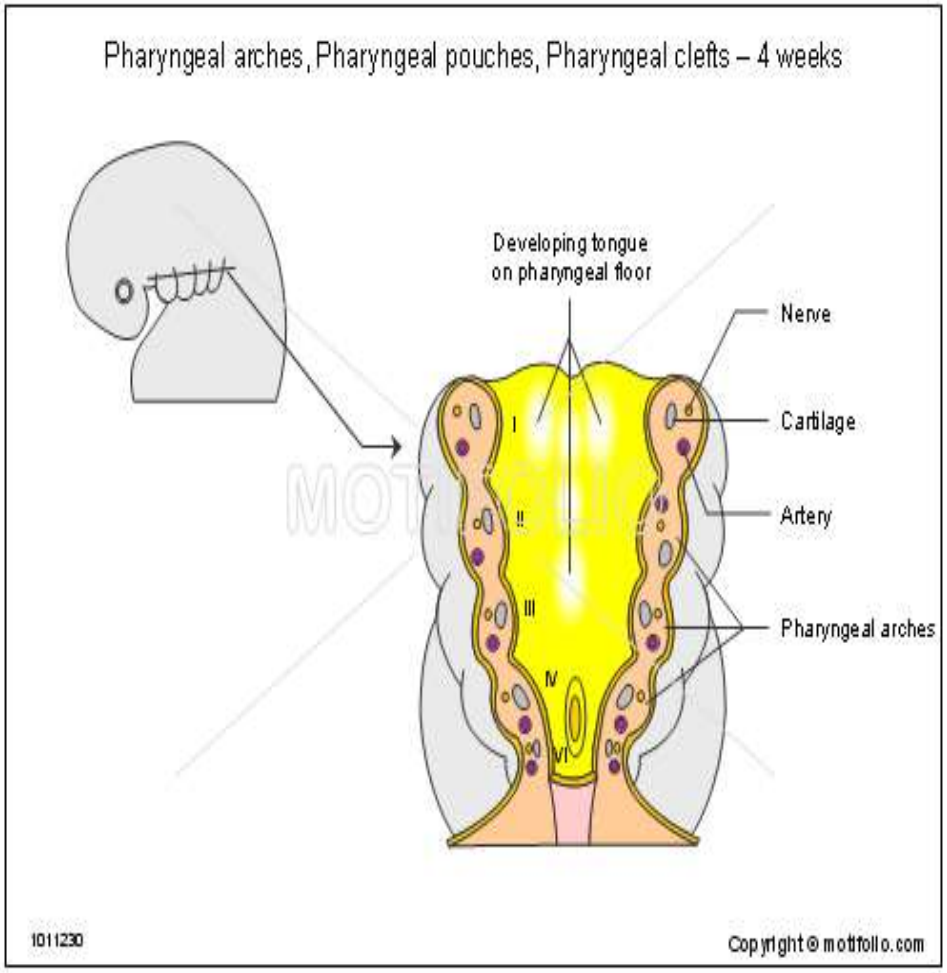
3- Microcephaly (small head)

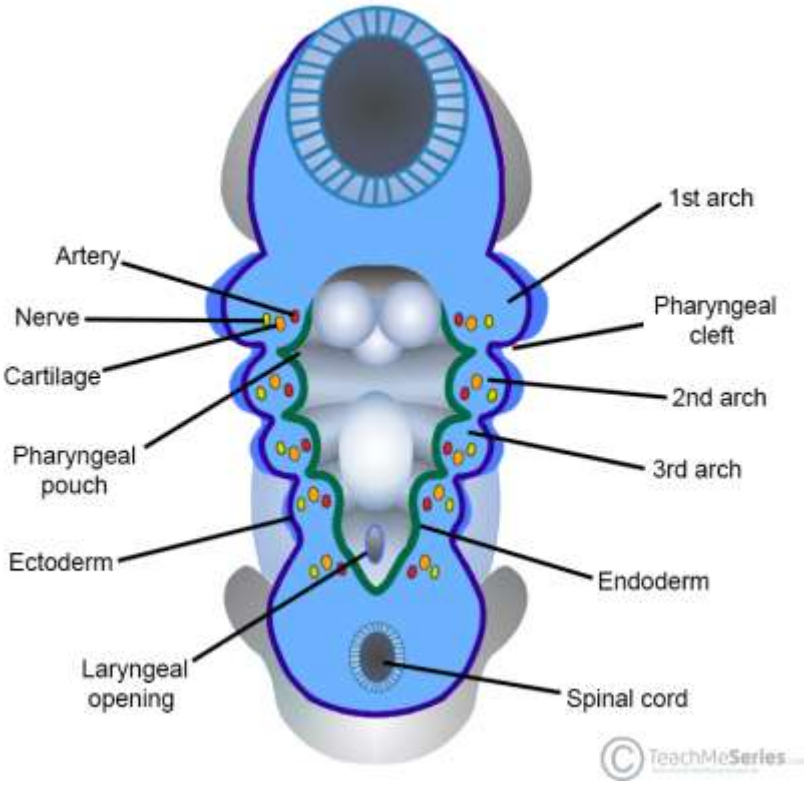
The brain does not develop properly resulting in a smaller than normal head.



pharyngeal arches

The development of the head and neck begins in the 4th and 5th week. Growth of mesenchymal tissue (connective tissue) in the cranial region of the embryo results in the formation of arches, separated by clefts. These are the **pharyngeal arches** and **pharyngeal clefts**.

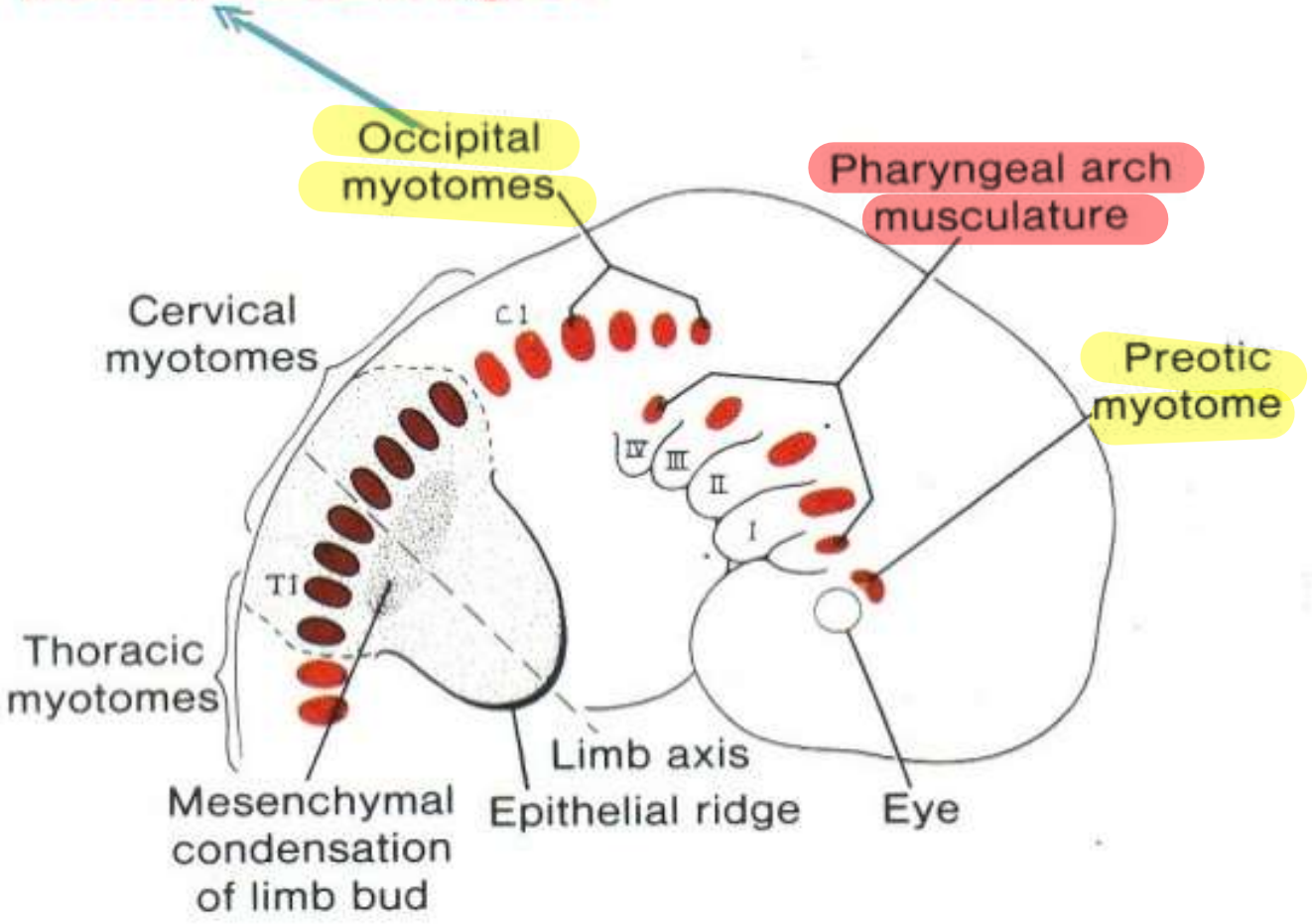




Development of the Face

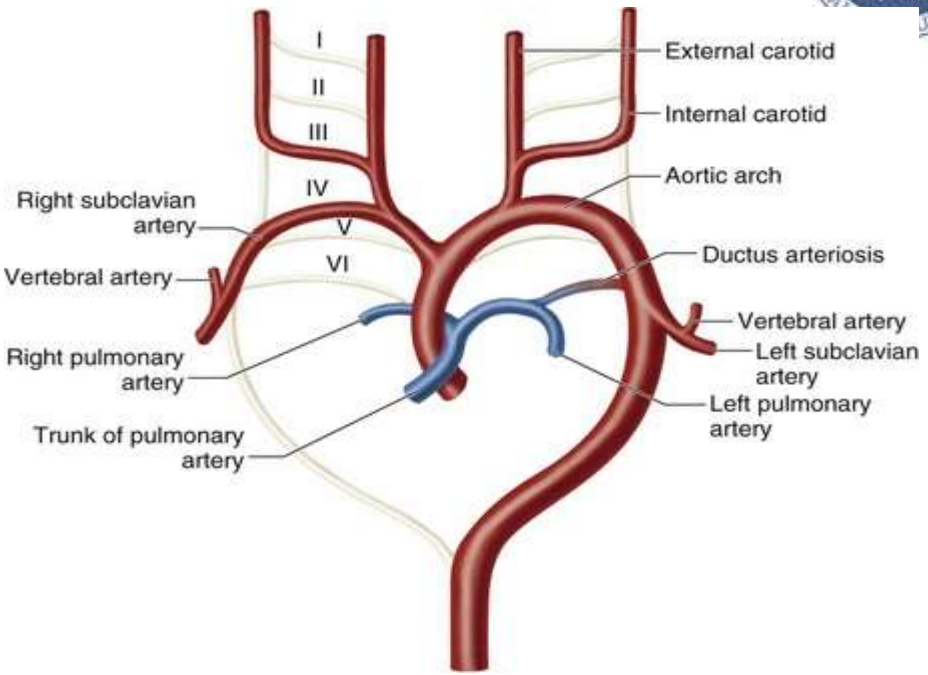
1. **Facial bones, muscles** ,are derived from the pharyngeal arches and supplied by the arteries and nerves of the arches.
2. **Tongue muscles** arise from occipital myotomes except the palatoglossus muscle and supplied by the hypoglossal nerve.
3. **The extra ocular muscles** arise from mesoderm surrounding the prochordal plate which form 3 preotic myotomes.

A- Muscles of tongue



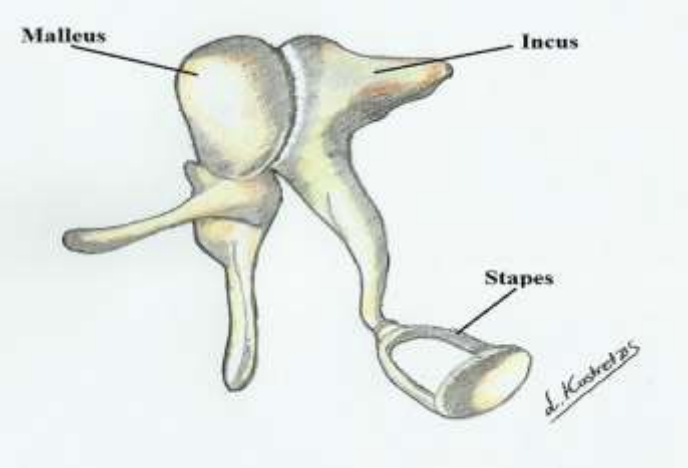
Arteries of the face: from the aortic arches

- 1st arch → disappears leaving the maxillary A.
- 2nd arch → disappears leaving the stapedial A.
- 3rd arch → Common carotid and proximal part of internal carotid.
  - The external carotid arises from the 3rd arch
- 4th arch → Rt. → part of subclavian Lt → part of arch of aorta.

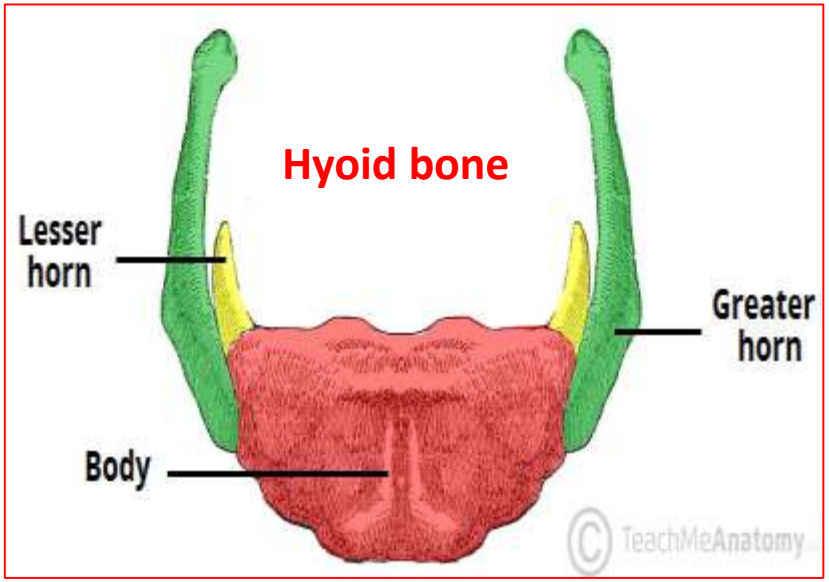


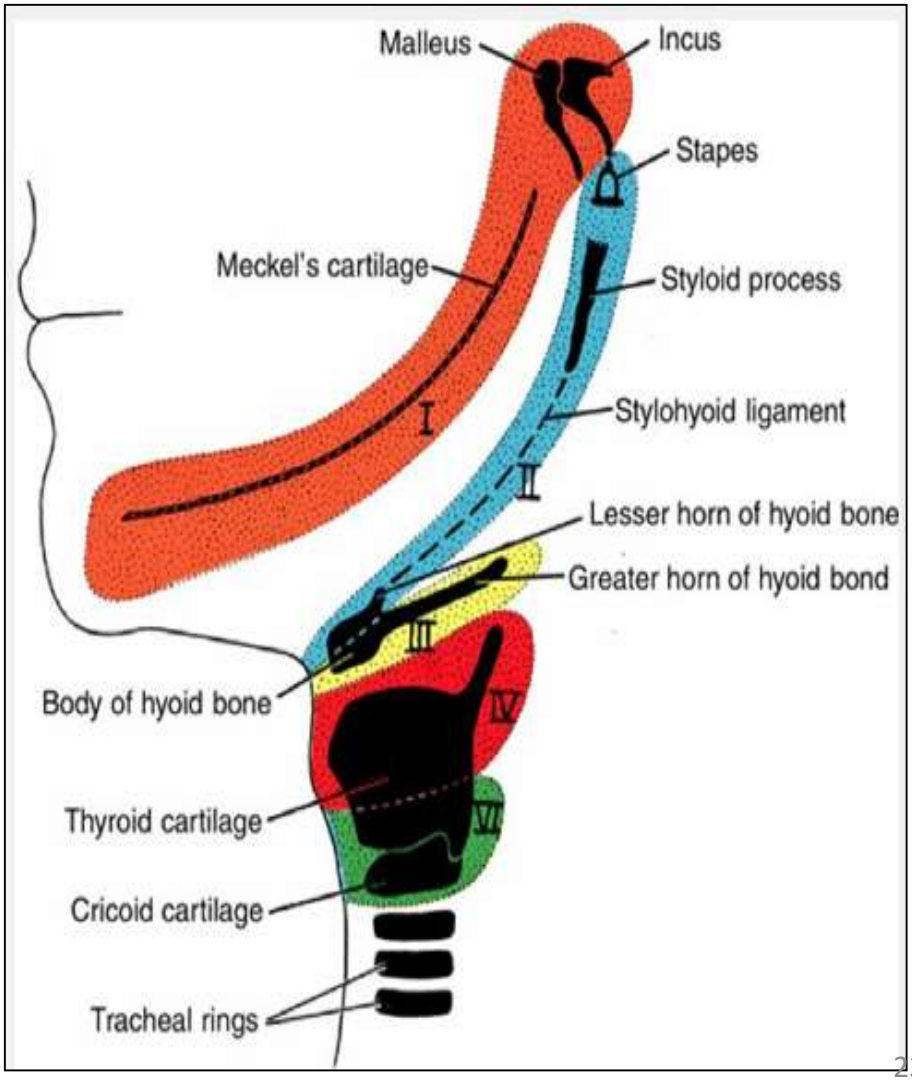
6th arch → proximal part on right and left sides gives pulmonary a.
Distal part: Rt. → disappears and Lt. → ductus arteriosus.

- Definitive structures formed by cartilaginous components of various pharyngeal arches



Bony ossicles





Fate of Pharyngeal Pouches

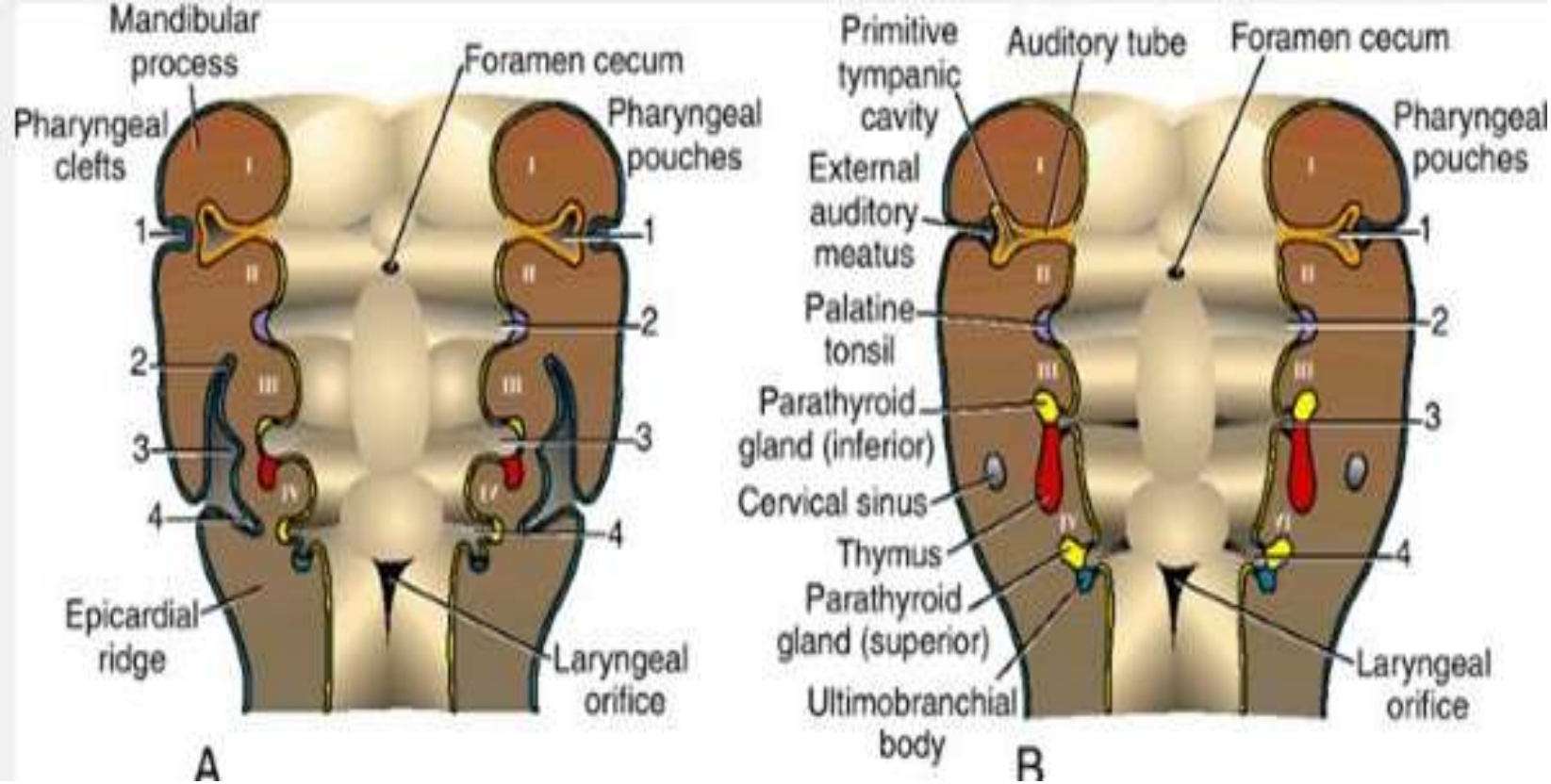
1. **Pharyngeal pouches:**
   - 1st pouch → tympanic cavity, mastoid antrum, inner surface of tympanic membrane and auditory (Eustachian) tube.
   - 2nd pouch → epithelium of palatine tonsils.
   - 3rd pouch → inferior parathyroid and thymus glands.
   - 4th pouch → superior parathyroid and ultimobranchial body which forms the parafollicular cells of thyroid gland

Fate of Pharyngeal Clefts

2. Pharyngeal clefts:

- 1st cleft → external auditory meatus and outer surface of the tympanic membrane.
- 2nd , 3rd and 4th fused forming the cervical sinus which buried under the surface ectoderm and finally disappears.
- N.B: the auricle develops from 6 elevations around the external auditory meatus and later fuse together.

Fate of Pharyngeal Pouches and Clefts



A- Are the pharyngeal clefts and formation of the cervical sinus which obliterate B- structures arises from the pouches

Congenital Anomalies of Pharyngeal Arches and Clefts.

- First arch syndrome: develop due to insufficient migration of neural crest cell into the first pharyngeal arch

- Treacher Collins:
  - Under developed zygomatic bone, deformed external ear
  - Defects of lower eye lid

- Pierre Robin syndrome:
  - Hypoplasia of mandible cleft palate, eye and ear defects





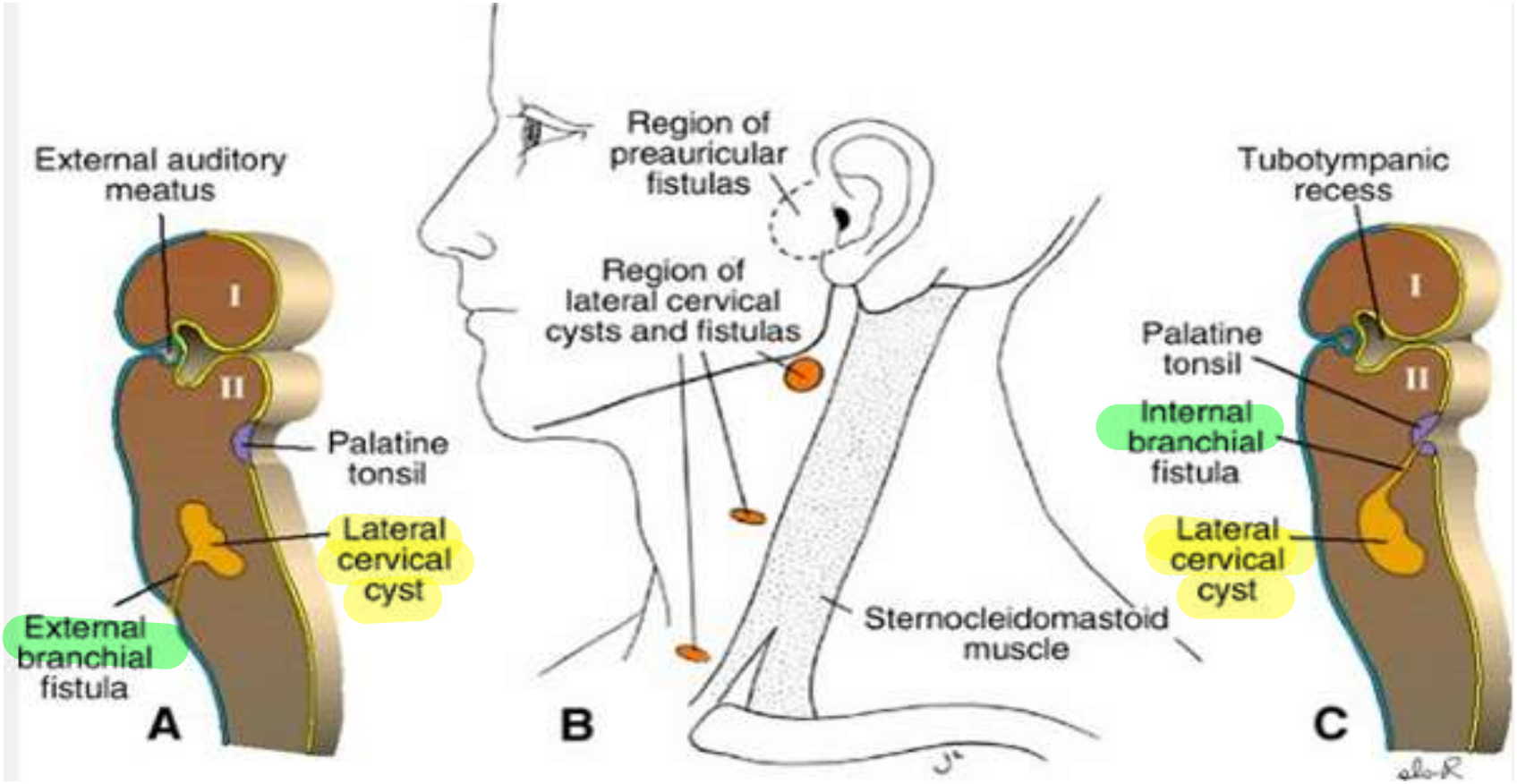
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Congenital Anomalies of Pharyngeal Pouches and Clefts.

2- Anomalies of the clefts:

1. Preauricular sinus.
2. Branchial cyst.
3. Branchial fistula which might be:
    1. Internal
    2. External

Congenital Anomalies of the Clefts



Lateral cervical cyst and fistula

Development of the Face

- The face develops from 5 swellings (prominences) appears during the 4th week around the stomodium (primitive mouth) due to proliferation of neural crest
  - Derived mesenchyme under the surface ectoderm.

1-One frontonasal prominence forms:

a) Frontal prominence → fore head + dorsum of nose(ala)

b) Nasal prominence → :

1. Lateral nasal prominence → side of nose
2. Medial nasal prominence → nasal septum & intermaxillary segment → 1- Philtrum of upper lip

   2. Part of upper jaw carrying the 4 incisor teeth.

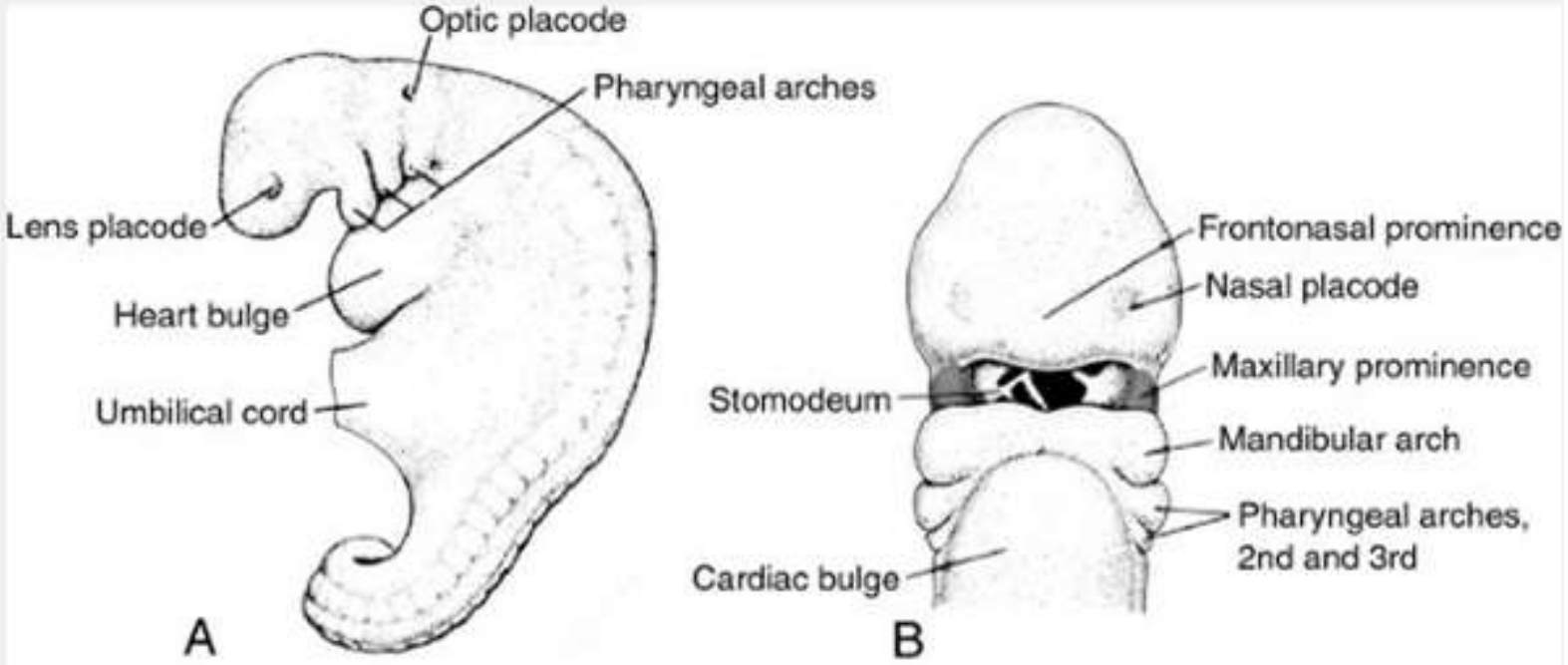
   3. Premaxilla (1ry palate)

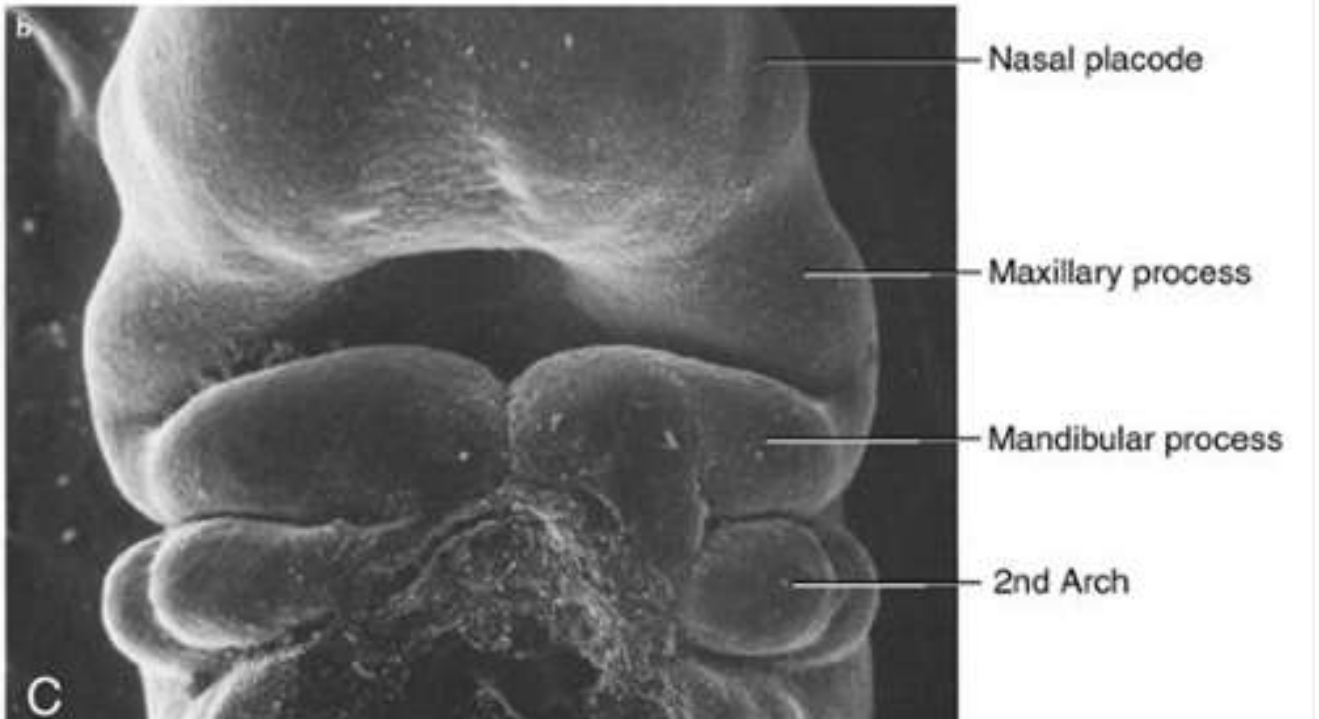
Development of the Face

2. Two Maxillary processes, from the first pharyngeal arch → the upper cheek + lateral part of upper lip.

3. Two Mandibular processes, from the first pharyngeal arch → chin+ lower lip+lower cheeks

- N.B: the ***PALATE*** develops by fusion of **primary palate** developed from the intermaxillary segment, and the **secondary palate** derived from the deep aspect of maxillary prominences.





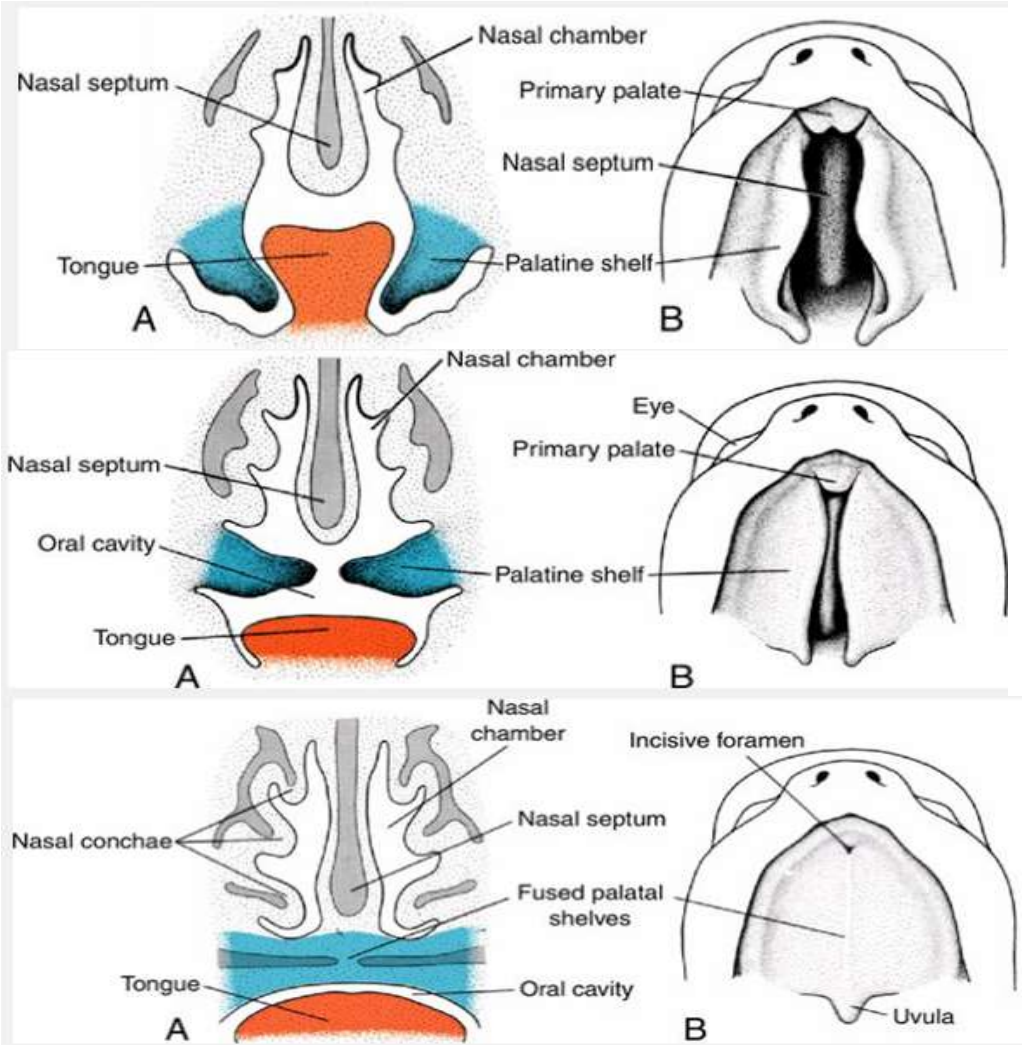
Frontonasal, maxillary and mandibular processes

Development of the Face.
(Palate)

Stages of the palate development .

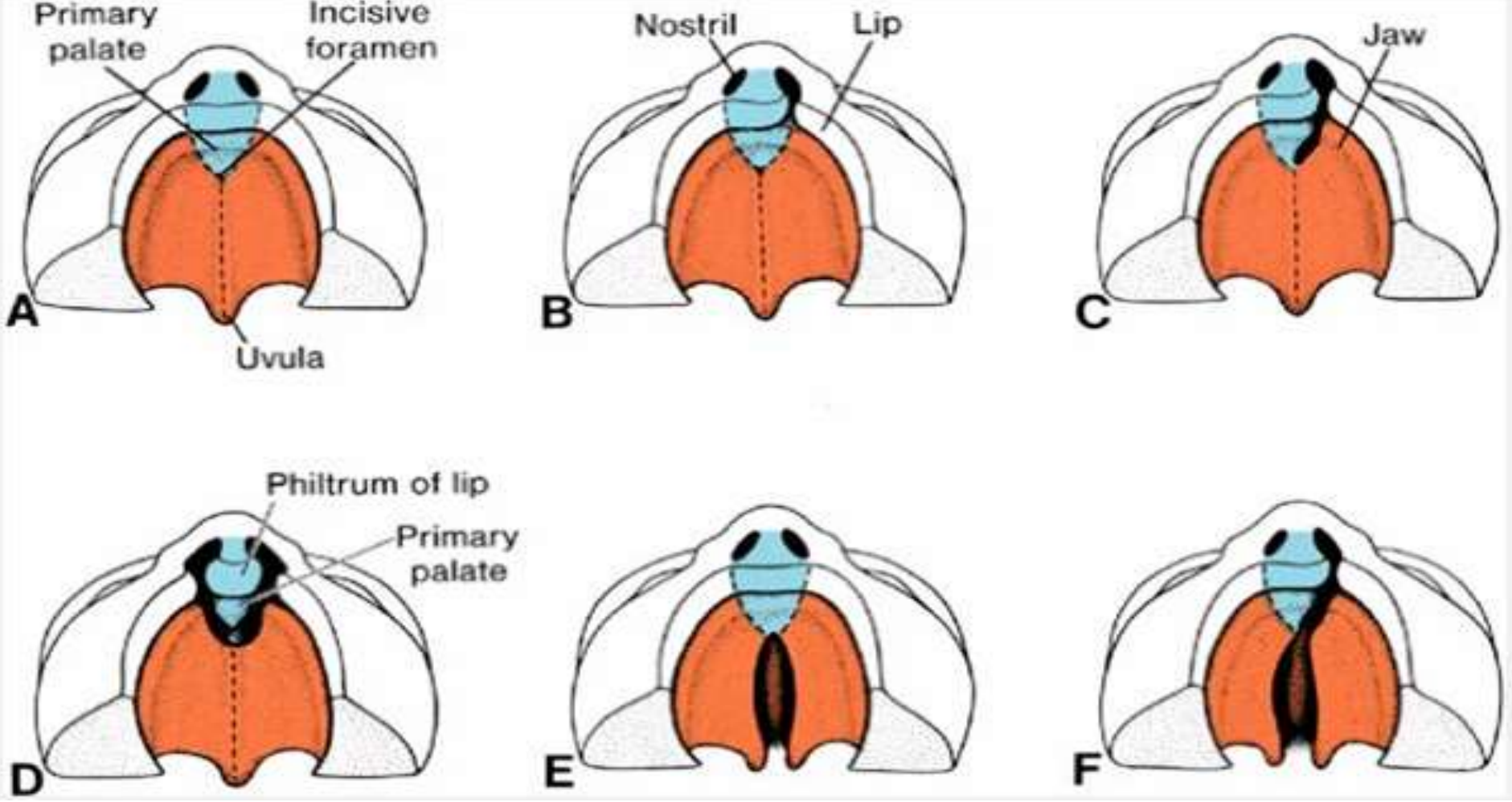
A – frontal view.

B- vertical view.



Congenital Anomalies of the Face. (Facial Clefts)

1. Cleft lip.(hare lip) : unilateral or bilateral. شفة مشقوقة
2. Oblique facial cleft. شق وجه
3. Cleft palate: which might be:
   1. Posterior to incisive foramen.
   2. Anterior to incisive foramen.
   3. Combination of both.
4. Rare midline anomalies.
5. Macrostomia.or microstomia فم كبير صغير
6. Dermoid cyst.



Various types of cleft lip and palate .A) normal B) unilateral cleft lip, C) unilateral cleft lip and palate. D)bilateral cleft lip and palate. E) isolated cleft palate. F) unilateral cleft lip and palate.











A) Incomplete cleft lip. B) Bilateral cleft lip. C) Cleft lip, cleft jaw and cleft palate. D) Isolated cleft palate. E) Oblique facial cleft

Summary

- Pharyngeal arches has its own artery, cranial nerve, muscle element and cartilage or skeletal element.
- Pharyngeal pouches gives rise to:
  - Middle ear cavity and auditory tube(pouch1)
  - Stroma of palatine tonsils(pouch2)
  - The inferior parathyroid and thymus glands(pouch3)
  - The superior parathyroid and ultimobranchial body (pouch 4 and 5)
- Pharyngeal clefts give rise to only one structure, the external auditor tube

Summary

- The two maxillary prominences with the two medial nasal prominences forms the upper lip.
- The intermaxillary segment is composed of

  a. The philtrum b. The upper jaw component

  c. The palatal component.
- The nose is derived from

  a. The frontonasal prominence b. The medial nasal prominences c. The lateral nasal prominence.
- The hard and soft palates are derived by the fusion of the palatal shelves of the maxillary prominences.